

NPDES Reasonable Potential (RP) Report

Report Generated: July 31, 2026

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1 Summary

Permit: PR0001031

Facility: PUERTO RICO ELECTRIC PWR AUTH

Date range: 2021-07-31 to 2026-07-31

Selected forms: 2C, 2F

Dilution ratio: 1

Confidence level: 0.95

Target percentile: 0.95

Hardness setting: Range

WARNING — PRELIMINARY REPORT (Quick Run)

This report was generated using Quick Run. The following parameters had unresolved data issues and were **automatically excluded** from analysis. Results should be considered preliminary. Re-run with all parameters reviewed for a final report.

Excluded parameters: Temperature; water deg. centigrade; Solids; settleable; Flow; in conduit or thru treatment plant; BOD; 5-day; 20 deg. C; Solids; total suspended; Oil & Grease; Chromium; total [as Cr]; Iron; total [as Fe]; Oxygen demand; chem. [high level] [COD]; Chlorine; free available; Nitrogen; Total

1.1 Included Pollutants

The following table shows the available data for each pollutant included in this analysis. The *Basis for Inclusion* column records why each pollutant is present: required by a selected NPDES form (2C, 2F); carried in because prior DMR data exists even though no selected form requires it (*Existing DMR*); or present only because the user supplied the monitoring data (*Manually Added*). A ‡ marker flags parameters whose record

includes monitoring data manually added by the user on top of the base DMR record. For a complete list of pollutants that had no associated data, refer to Section 5.

Pollutant	Basis for Inclusion	Ref Param	# Samples
Arsenic	Form 2C	01002	2
Cadmium	Form 2C	01027	2
Chlorine (total residual, TRC)	Existing DMR	50060	100
Chromium, hexavalent [as Cr]	Existing DMR	01032	2
Copper	Form 2C	01042	167
Cyanide	Form 2C	51173	109
Dissolved oxygen	Existing DMR	00300	112
Mercury	Form 2C	71900	52
Nickel	Form 2C	01067	5
Pentachlorophenol	Form 2C	39032	4
Polychlorinated biphenyls [PCBs]	Existing DMR	39516	8
Silver	Form 2C	01077	58
Sulfates, total [as SO ₄]	Existing DMR	51865	2
Sulfide-hydrogen sulfide [undissociated]	Existing DMR	51202	53
Thallium	Form 2C	01059	56
Turbidity	Existing DMR	00070	111
Zinc	Form 2C	01092	87
pH (maximum)	Forms 2C & 2F	00400	259
pH (minimum)	Forms 2C & 2F	00400	259

[†] WQS criterion manually entered by user; see Manual WQS Overrides section.

[‡] Includes monitoring data manually added by the user.

2 RP Findings

RP column: **YES** indicates the assessed value exceeds the criterion. **DEPENDS** indicates an exceedance was not observed but the tool cannot conclude no reasonable potential — see the note on the pollutant's section. **NO** indicates no exceedance.

The Basis column distinguishes a *Calculated* receiving water concentration (TSD multiplying-factor projection, dilution ratio applied) from an *Observed* maximum or minimum compared directly to the criterion with no projection and no dilution. Values rounded to 2 decimals.

WQS ID	Pollutant	Outfall	Water Class	WQS Criterion	Value	Basis	RP
79230	Chlorine (total residual, TRC)	External Outfall 001	class SB waters	7.50	210.41	Calculated	YES
79230	Chlorine (total residual, TRC)	External Outfall 01A	class SB waters	7.50	2602.61	Calculated	YES
79230	Chlorine (total residual, TRC)	External Outfall 01C	class SB waters	7.50	217.01	Calculated	YES
79235	Copper	External Outfall 001	class SB waters	3.73	15.57	Calculated	YES
79232	Cyanide	External Outfall 001	class SB waters	1.00	2.90	Calculated	YES
79590	Dissolved Oxygen	External Outfall 001	class SB waters	5.00	4.40	Observed min	YES

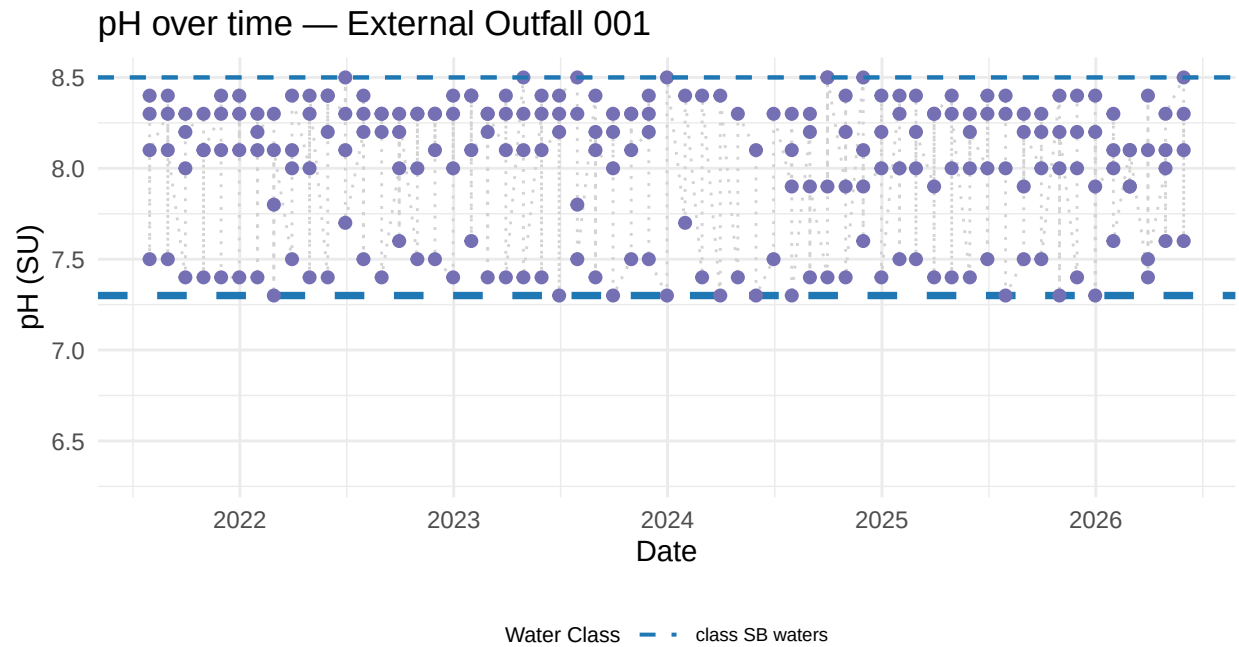
(continued)

WQS ID	Pollutant	Outfall	Water Class	WQS Criterion	Value	Basis	RP
79244	Mercury	External Outfall 001	class SB waters	0.05	0.10	Calculated	YES
79247	Nickel	External Outfall 001	class SB waters	8.28	100.00	Calculated	YES
79247	Nickel	External Outfall 002	class SB waters	8.28	74.37	Calculated	YES
79401	Polychlorinated Biphenyls (PCBs)	External Outfall 001	class SB waters	0.00	0.95	Calculated	YES
79597-51865	Sulfates, total [as SO ₄]	External Outfall 001	class SB waters	2800.00	6435.52	Calculated	YES
79263	Zinc	External Outfall 01A	class SB waters	85.62	1740.73	Calculated	YES
79263	Zinc	External Outfall 01C	class SB waters	85.62	202.56	Calculated	YES
79595	pH (maximum)	External Outfall 002	class SB waters	8.50	8.90	Observed max	YES
79595	pH (maximum)	External Outfall 01A	class SB waters	8.50	8.90	Observed max	YES
79595	pH (maximum)	External Outfall 01C	class SB waters	8.50	8.80	Observed max	YES
79595	pH (minimum)	External Outfall 01A	class SB waters	7.30	6.40	Observed min	YES
79590	Dissolved Oxygen	External Outfall 002	class SB waters	5.00	9.20	Observed min	DE-PENDS
79224	Arsenic	External Outfall 001	class SB waters	36.00	6.45	Calculated	NO
79227	Cadmium	External Outfall 001	class SB waters	7.95	0.27	Calculated	NO
79239	Chromium, hexavalent [as Cr]	External Outfall 001	class SB waters	50.40	0.00	Calculated	NO
79323	Pentachlorophenol	External Outfall 001	class SB waters	0.40	0.00	Calculated	NO
79250	Silver	External Outfall 001	class SB waters	2.24	0.91	Calculated	NO
79258	Sulfide-hydrogen sulfide [undissociated]	External Outfall 001	class SB waters	2.00	1.17	Calculated	NO
79260	Thallium	External Outfall 001	class SB waters	0.47	0.42	Calculated	NO
79596	Turbidity	External Outfall 001	class SB waters	10.00	10.00	Observed max	NO
79595	pH (maximum)	External Outfall 001	class SB waters	8.50	8.50	Observed max	NO
79595	pH (minimum)	External Outfall 001	class SB waters	7.30	7.30	Observed min	NO
79595	pH (minimum)	External Outfall 002	class SB waters	7.30	8.90	Observed min	NO
79595	pH (minimum)	External Outfall 01C	class SB waters	7.30	7.30	Observed min	NO

3 Non-Concentration Based Measures

3.1 External Outfall 001 — pH

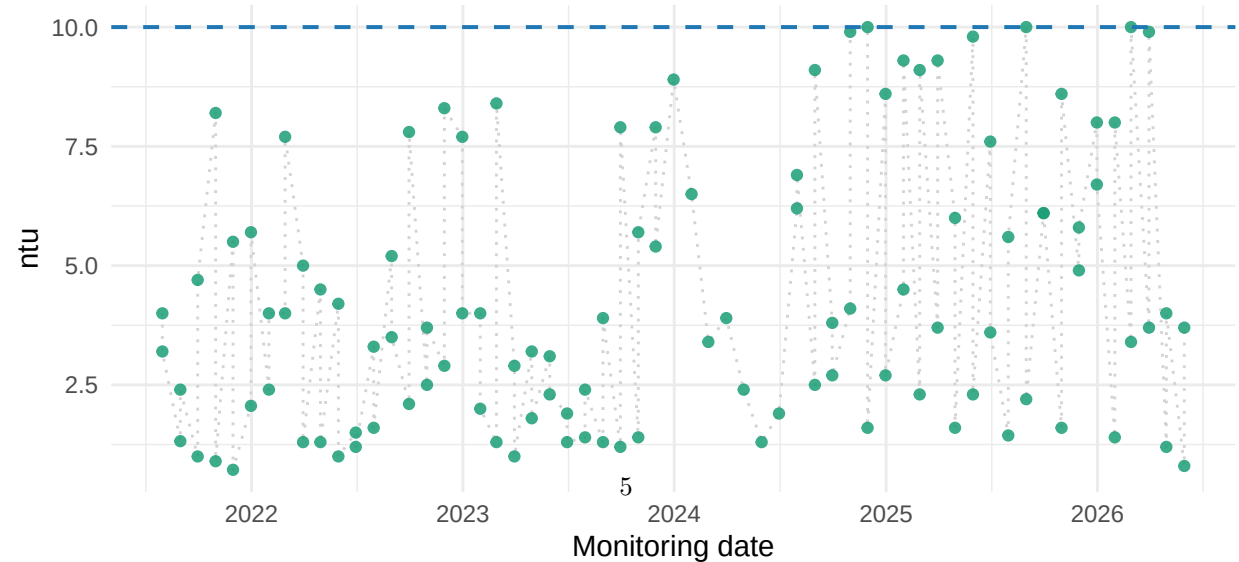
Max pH	Min pH
8.5	7.3



WQS ID	Water Class	Bound	WQS Criterion	Observed pH	RP
79595	class SB waters	Max	8.5	8.5	NO
79595	class SB waters	Min	7.3	7.3	NO

3.1.1 Turbidity — External Outfall 001

WQS ID	Water Class	WQS Criterion	RP	Observed Max
79596	class SB waters	10	NO	10



WQS ID	Water Class	WQS Criterion	RP	Observed Min
79590	class SB waters	5	YES	4.4

WQS ID	Water Class	Bound	WQS Criterion	Observed pH	RP
79595	class SB waters	Max	8.5	8.9	YES
79595	class SB waters	Min	7.3	8.9	NO

WQS ID	Water Class	WQS Criterion	RP	Observed Min
79590	class SB waters	5	DEPENDS	9.2

WQS ID	Water Class	Bound	WQS Criterion	Observed pH	RP
79595	class SB waters	Max	8.5	8.9	YES
79595	class SB waters	Min	7.3	6.4	YES

WQS ID	Water Class	Bound	WQS Criterion	Observed pH	RP
79595	class SB waters	Max	8.5	8.8	YES
79595	class SB waters	Min	7.3	7.3	NO

4 Concentration Based Measures

4.1 RP Equations

The following equations estimate the Receiving Water Concentration (RWC) for parameters assessed by projection. They do **not** apply to parameters assessed by direct comparison — pH, temperature, dissolved oxygen, Enterococci, color, turbidity and oil & grease — whose sections report an observed maximum or minimum instead, with no multiplying factor and no dilution ratio. Each such section states this explicitly.

$$\begin{aligned}\sigma_g &= \sqrt{\ln(CV^2 + 1)} \\ p &= (1 - CL)^{1/n} \\ z_p &= \Phi^{-1}(p), \quad z_{\text{target}} = \Phi^{-1}(\text{target}) \\ MF &= \exp((z_{\text{target}} - z_p) \sigma_g)\end{aligned}$$

$$RWC = \frac{\max(x) \times MF}{DR}$$

4.2 Arsenic

Reason Included: Form(s) 2C selected.

4.2.1 External Outfall 001

- Samples: 2
- Units converted: No

RP Breakdown by Water Class

RWC Calculation

WQS ID	Water Class	WQS Criterion	RP
79224	class SB waters	36	NO

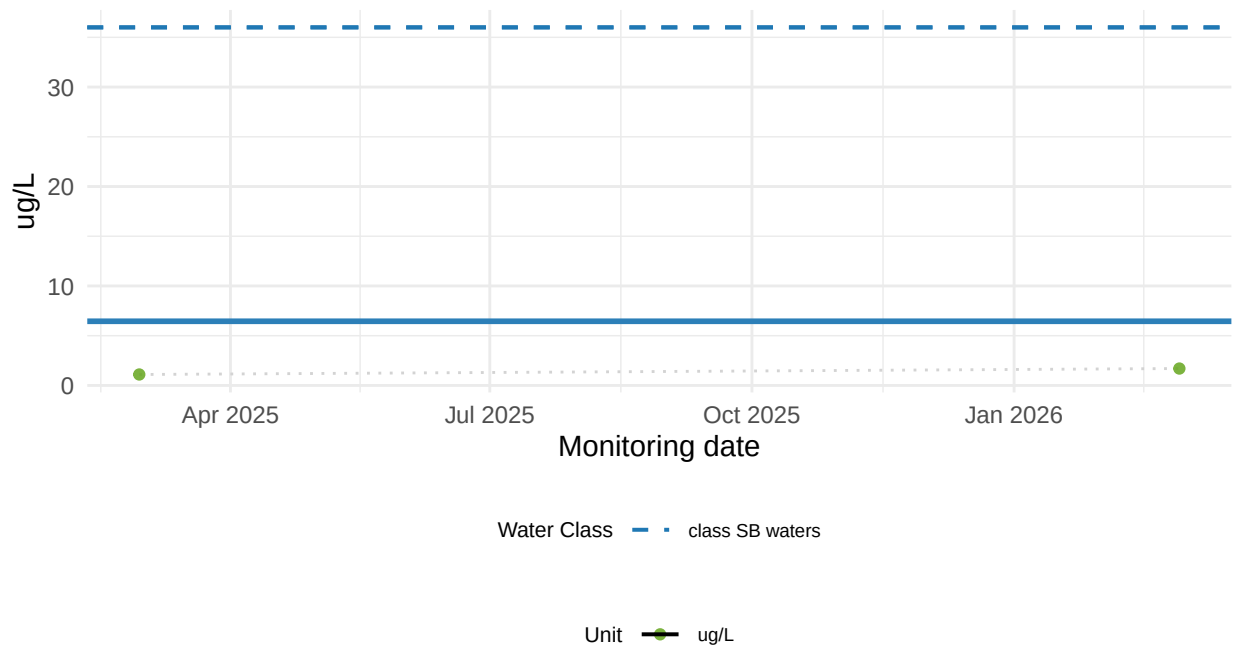
$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/2} = 0.22$$

$$z_p = -0.76, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - -0.76) \times 0.55) = 3.8$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{1.7 \times 3.8}{1} = 6.5$$



WQS ID	Water Class	WQS Criterion	RWC	RP
79224	class SB waters	36	6.450702	NO

4.3 Cadmium

Reason Included: Form(s) 2C selected.

4.3.1 External Outfall 001

- Samples: 2
- Units converted: No

RP Breakdown by Water Class

RWC Calculation

WQS ID	Water Class	WQS Criterion	RP
79227	class SB waters	7.95	NO

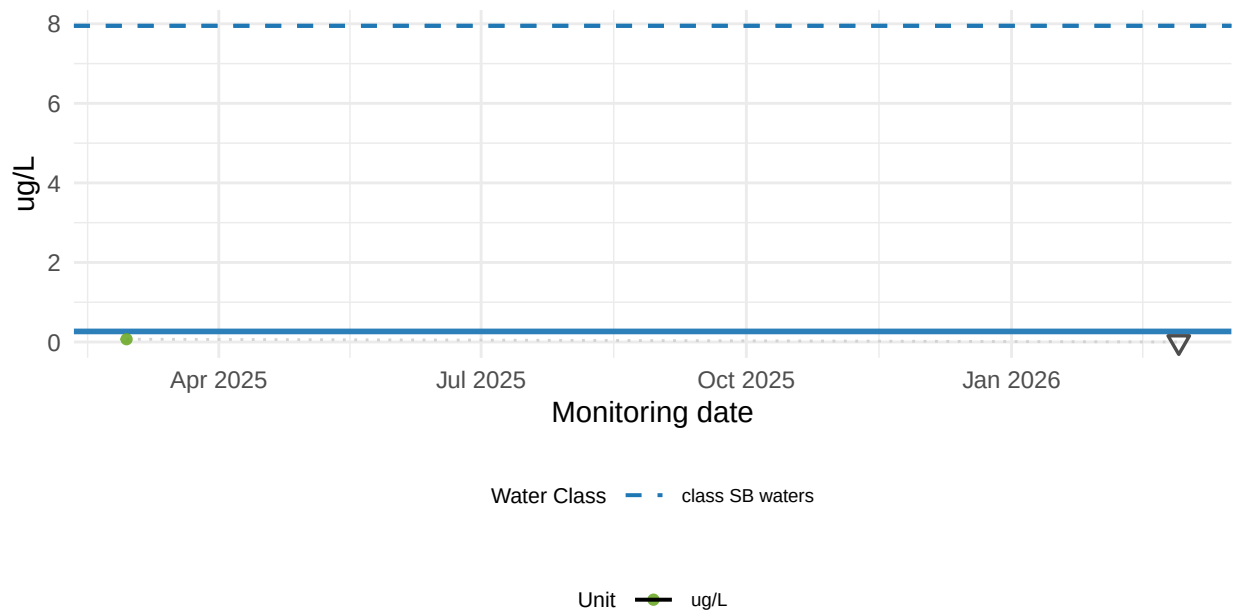
$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/2} = 0.22$$

$$z_p = -0.76, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - -0.76) \times 0.55) = 3.8$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0.07 \times 3.8}{1} = 0.27$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79227	class SB waters	7.95	0.2656171	NO

4.4 Copper

Reason Included: Form(s) 2C selected.

4.4.1 External Outfall 001

- Samples: 109
- Units converted: No

RP Breakdown by Water Class

RWC Calculation

WQS ID	Water Class	WQS Criterion	RP
79235	class SB waters	3.73	YES

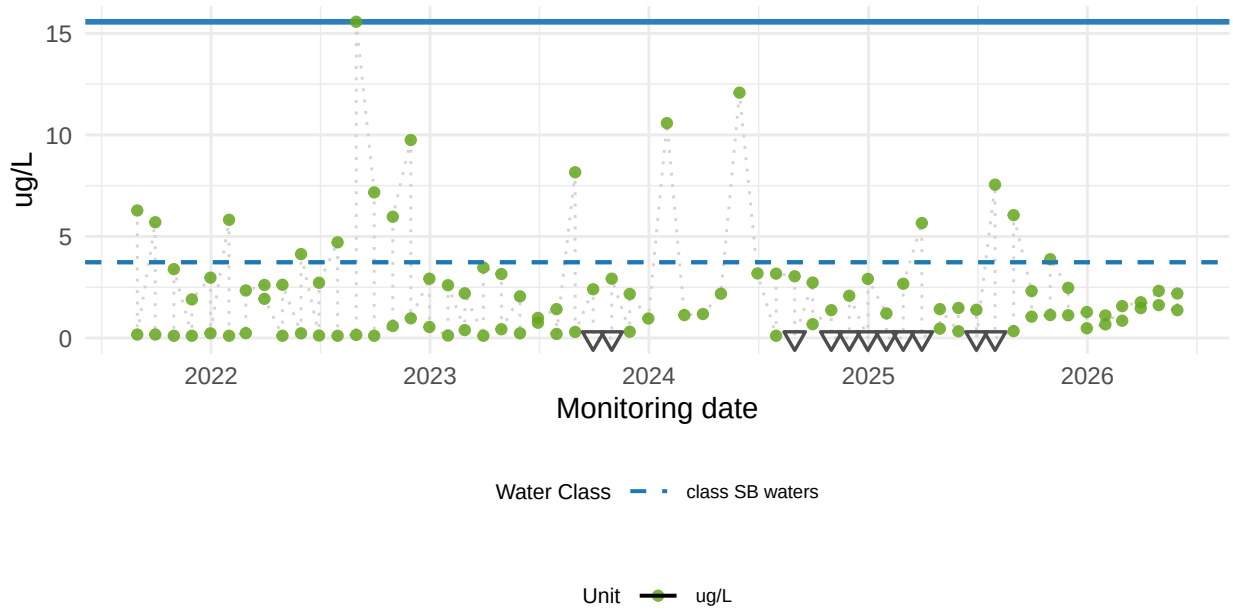
$$\sigma_g = \sqrt{\ln(1.3^2 + 1)} = 0.98$$

$$p = (1 - 0.95)^{1/109} = 0.97$$

$$z_p = 1.9, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.9) \times 0.98) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{16 \times 1}{1} = 16$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79235	class SB waters	3.73	15.57	YES

4.5 Cyanide

Reason Included: Form(s) 2C selected.

4.5.1 External Outfall 001

- Samples: 109
- Units converted: No

RP Breakdown by Water Class

RWC Calculation

WQS ID	Water Class	WQS Criterion	RP
79232	class SB waters	1	YES

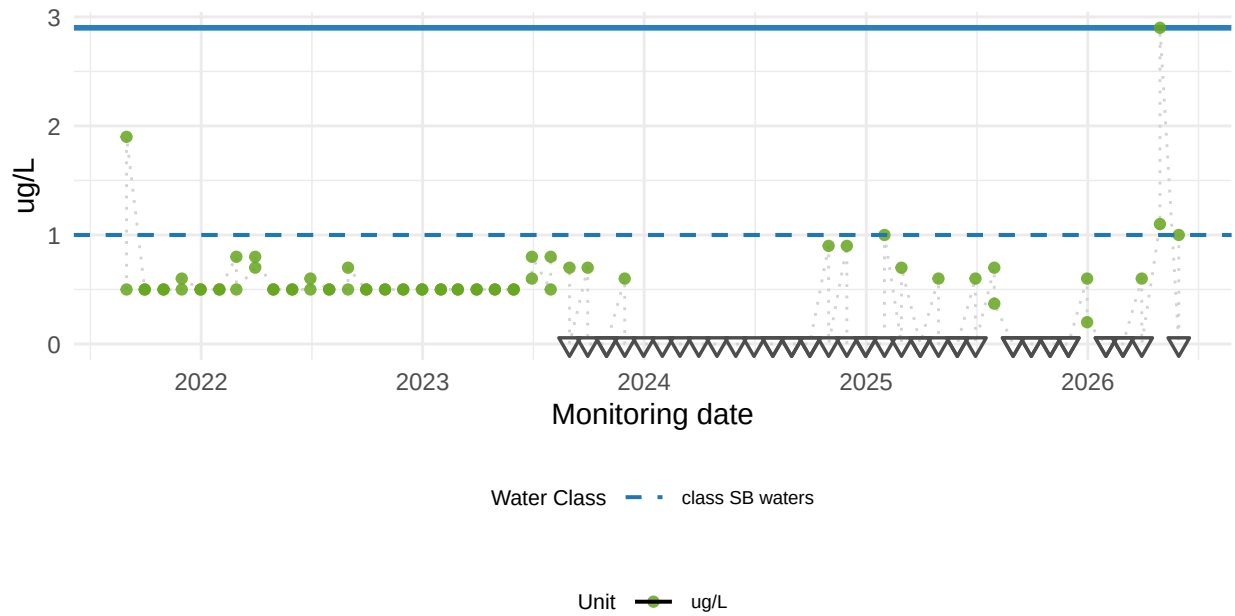
$$\sigma_g = \sqrt{\ln(1.1^2 + 1)} = 0.89$$

$$p = (1 - 0.95)^{1/109} = 0.97$$

$$z_p = 1.9, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.9) \times 0.89) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{2.9 \times 1}{1} = 2.9$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79232	class SB waters	1	2.9	YES

4.6 Mercury

Reason Included: Form(s) 2C selected.

4.6.1 External Outfall 001

- Samples: 52
- Units converted: No

RP Breakdown by Water Class

RWC Calculation

WQS ID	Water Class	WQS Criterion	RP
79244	class SB waters	0.051	YES

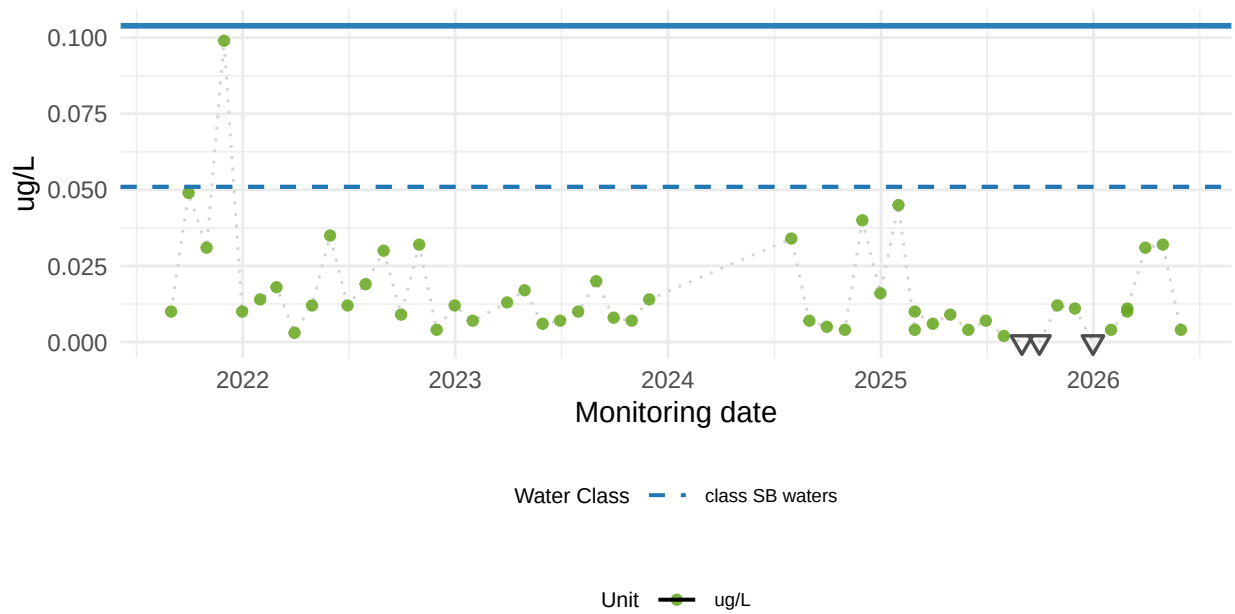
$$\sigma_g = \sqrt{\ln(1.1^2 + 1)} = 0.88$$

$$p = (1 - 0.95)^{1/52} = 0.94$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.6) \times 0.88) = 1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0.099 \times 1}{1} = 0.1$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79244	class SB waters	0.051	0.1039213	YES

4.7 Chlorine (total residual, TRC)

Reason Included: Not associated with selected forms — prior DMR data exists.

4.7.1 External Outfall 001

- Samples: 52
- Units converted: Yes

RP Breakdown by Water Class

RWC Calculation

WQS ID	Water Class	WQS Criterion	RP
79230	class SB waters	7.5	YES

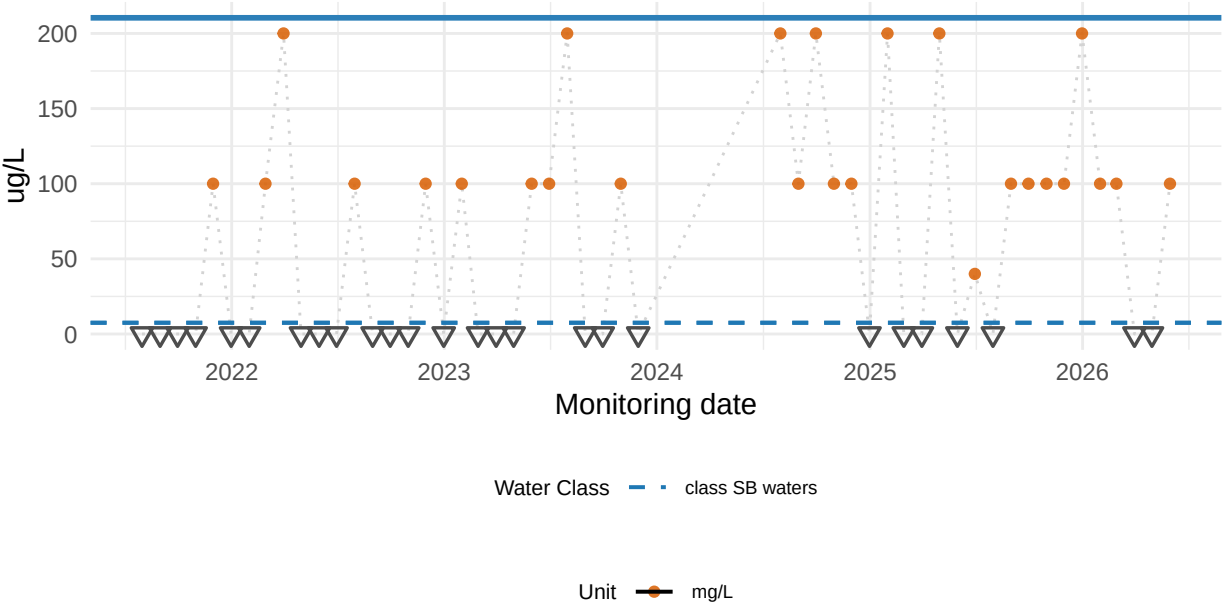
$$\sigma_g = \sqrt{\ln(1.1^2 + 1)} = 0.92$$

$$p = (1 - 0.95)^{1/52} = 0.94$$

$$z_p = 1.6, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.6) \times 0.92) = 1.1$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{200 \times 1.1}{1} = 210$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79230	class SB waters	7.5	210.41	YES

4.7.2 External Outfall 01A

- Samples: 30
- Units converted: Yes

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79230	class SB waters	7.5	YES

RWC Calculation

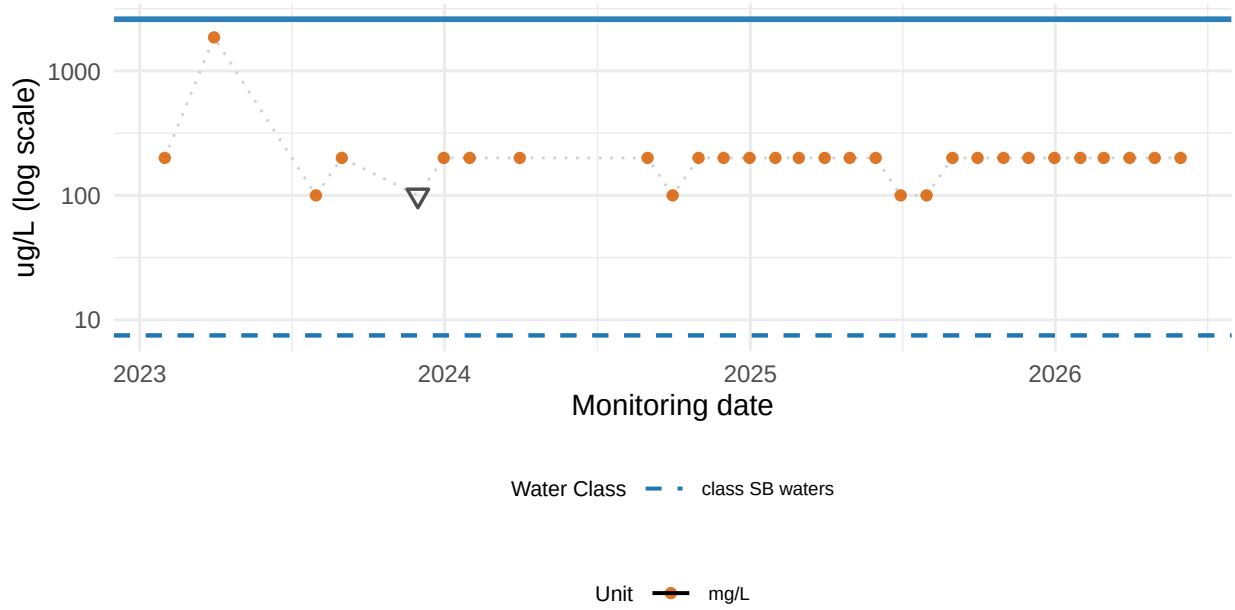
$$\sigma_g = \sqrt{\ln(1.3^2 + 1)} = 1$$

$$p = (1 - 0.95)^{1/30} = 0.9$$

$$z_p = 1.3, z_{\text{target}} = 1.6$$

$$\text{MF} = \exp((1.6 - 1.3) \times 1) = 1.4$$

$$\text{RWC} = \frac{\max(x) \times \text{MF}}{\text{DR}} = \frac{1900 \times 1.4}{1} = 2600$$



Open triangles are non-detects, plotted at 100.

WQS ID	Water Class	WQS Criterion	RWC	RP
79230	class SB waters	7.5	2602.611	YES

4.7.3 External Outfall 01C

- Samples: 18
- Units converted: Yes

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79230	class SB waters	7.5	YES

RWC Calculation

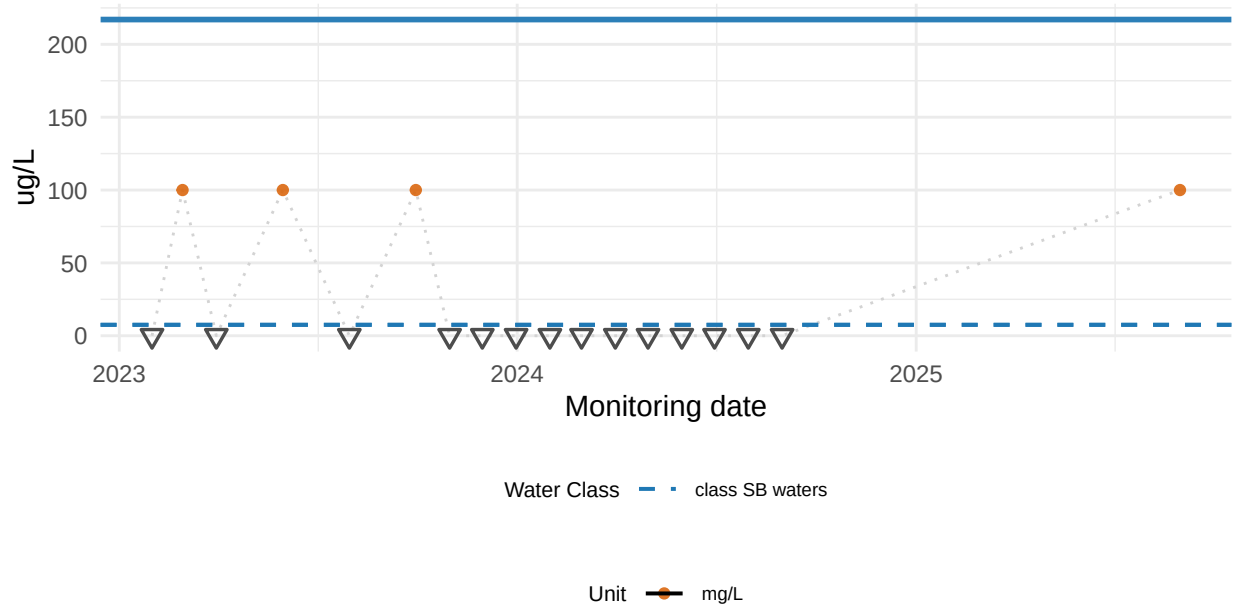
$$\sigma_g = \sqrt{\ln(1.9^2 + 1)} = 1.2$$

$$p = (1 - 0.95)^{1/18} = 0.85$$

$$z_p = 1, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1) \times 1.2) = 2.2$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{100 \times 2.2}{1} = 220$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79230	class SB waters	7.5	217.0085	YES

4.8 Chromium, hexavalent [as Cr]

Reason Included: Not associated with selected forms — prior DMR data exists.

4.8.1 External Outfall 001

- Samples: 2
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79239	class SB waters	50.4	NO

RWC Calculation

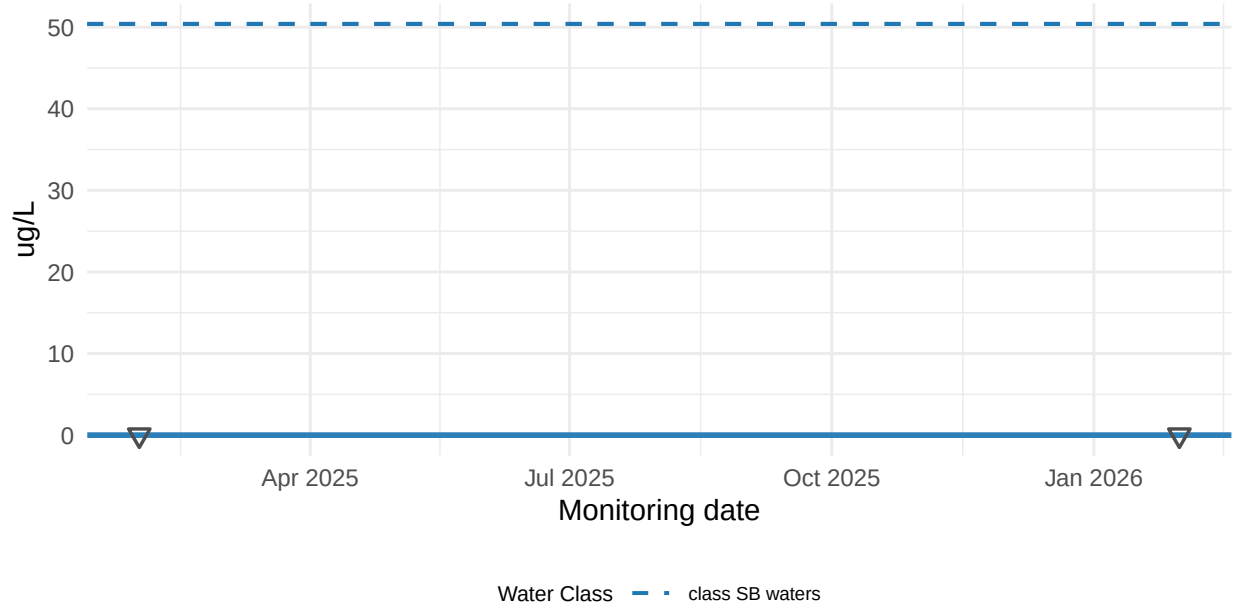
$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/2} = 0.22$$

$$z_p = -0.76, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - -0.76) \times 0.55) = 3.8$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0 \times 3.8}{1} = 0$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79239	class SB waters	50.4	0	NO

4.9 Polychlorinated Biphenyls (PCBs)

4.9.1 External Outfall 001

- Samples: 8
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79401	class SB waters	0.00064	YES

RWC Calculation

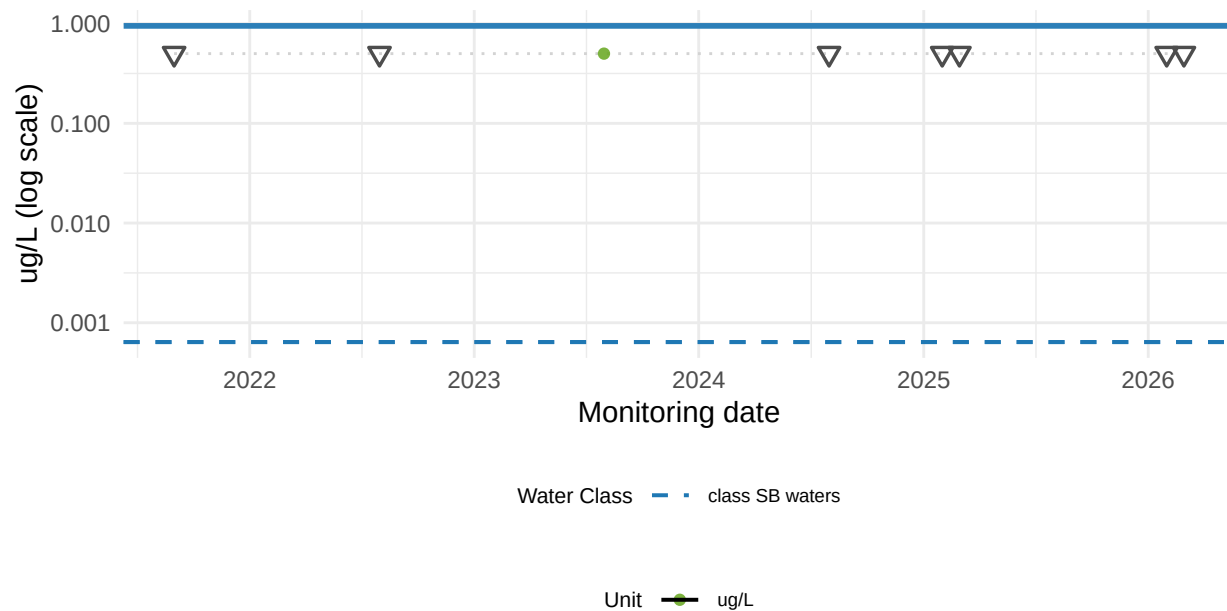
$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/8} = 0.69$$

$$z_p = 0.49, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 0.49) \times 0.55) = 1.9$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{0.5 \times 1.9}{1} = 0.95$$



Open triangles are non-detects, plotted at 0.5.

WQS ID	Water Class	WQS Criterion	RWC	RP
79401	class SB waters	0.00064	0.949014	YES

4.10 Sulfide-hydrogen sulfide [undissociated]

Reason Included: Not associated with selected forms — prior DMR data exists.

4.10.1 External Outfall 001

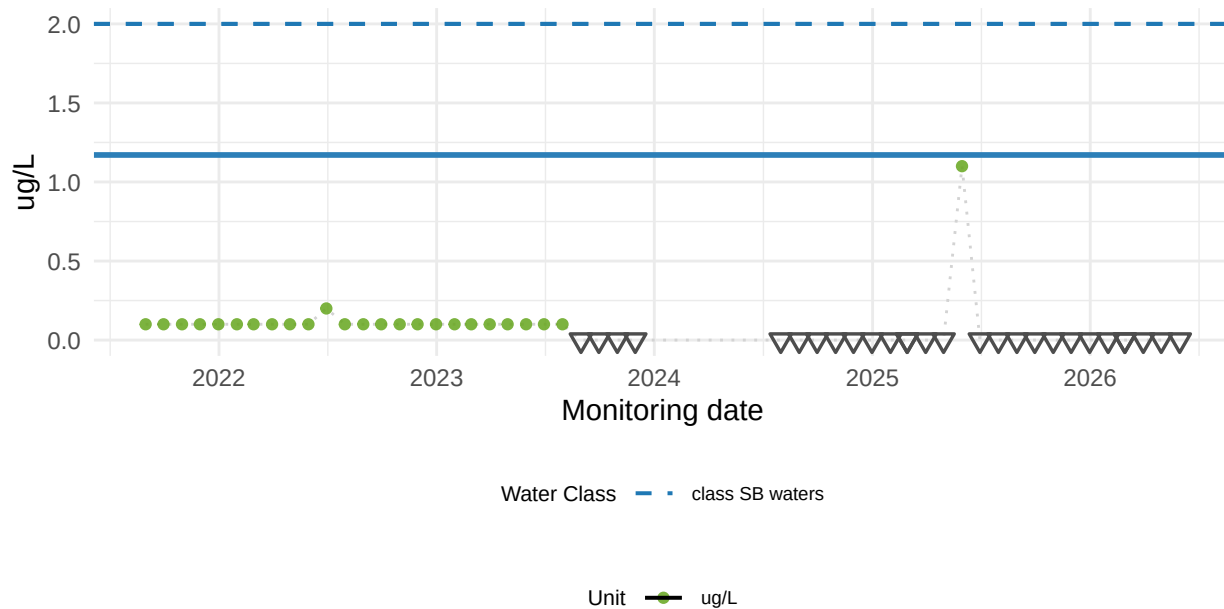
- Samples: 53
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79258	class SB waters	2	NO

RWC Calculation

$$\begin{aligned}\sigma_g &= \sqrt{\ln(2.3^2 + 1)} = 1.3 \\ p &= (1 - 0.95)^{1/53} = 0.95 \\ z_p &= 1.6, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - 1.6) \times 1.3) = 1.1 \\ RWC &= \frac{\max(x) \times MF}{DR} = \frac{1.1 \times 1.1}{1} = 1.2\end{aligned}$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79258	class SB waters	2	1.170764	NO

4.11 Nickel

Reason Included: Form(s) 2C selected.

4.11.1 External Outfall 001

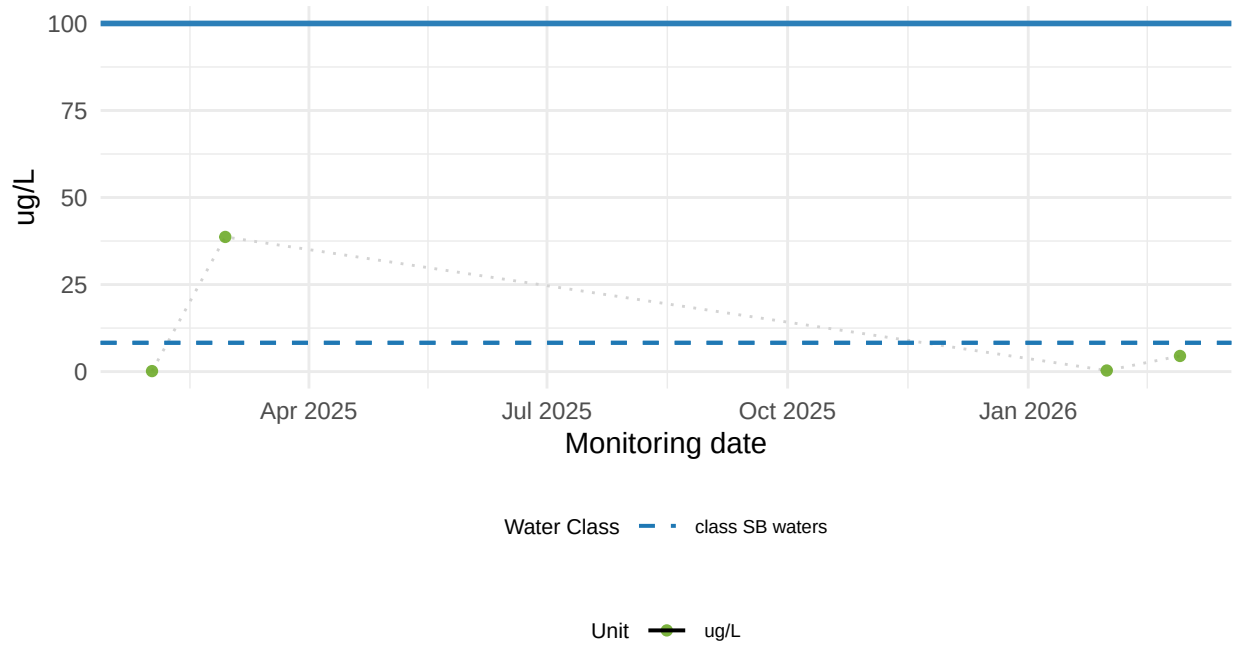
- Samples: 4
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79247	class SB waters	8.28	YES

RWC Calculation

$$\begin{aligned}\sigma_g &= \sqrt{\ln(0.6^2 + 1)} = 0.55 \\ p &= (1 - 0.95)^{1/4} = 0.47 \\ z_p &= -0.068, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - -0.068) \times 0.55) = 2.6 \\ RWC &= \frac{\max(x) \times MF}{DR} = \frac{39 \times 2.6}{1} = 100\end{aligned}$$



WQS ID	Water Class	WQS Criterion	RWC	RP
79247	class SB waters	8.28	99.99828	YES

4.11.2 External Outfall 002

- Samples: 1
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79247	class SB waters	8.28	YES

RWC Calculation

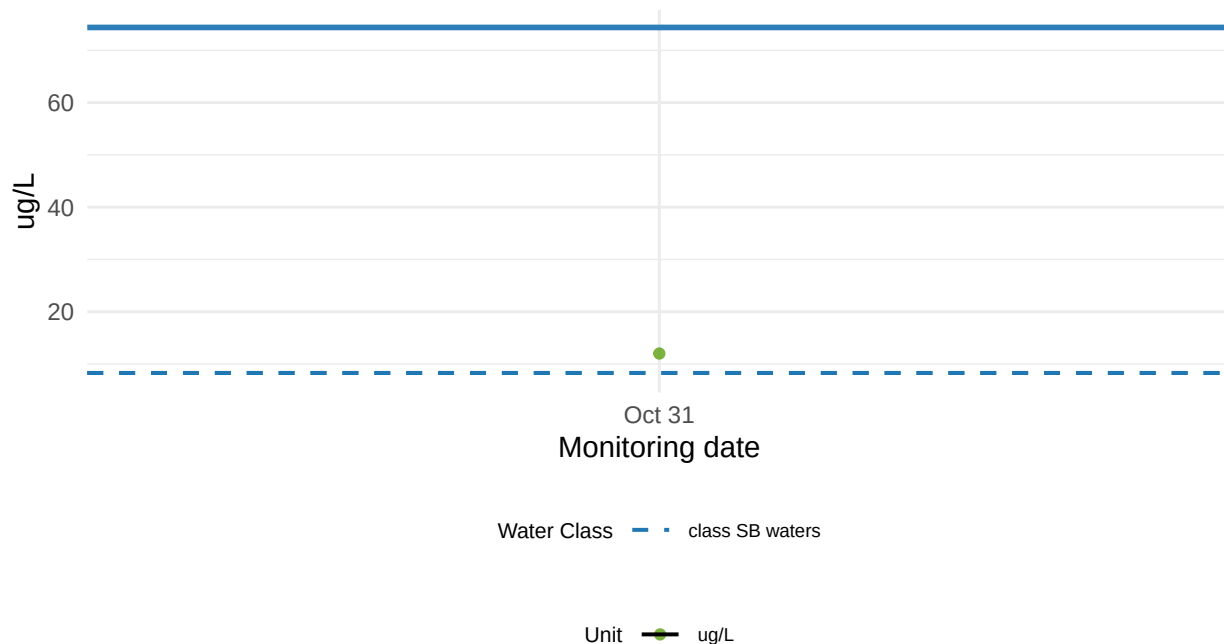
$$\sigma_g = \sqrt{\ln(0.6^2 + 1)} = 0.55$$

$$p = (1 - 0.95)^{1/1} = 0.05$$

$$z_p = -1.6, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - -1.6) \times 0.55) = 6.2$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{12 \times 6.2}{1} = 74$$



WQS ID	Water Class	WQS Criterion	RWC	RP
79247	class SB waters	8.28	74.37294	YES

4.12 Pentachlorophenol

Reason Included: Form(s) 2C selected.

4.12.1 External Outfall 001

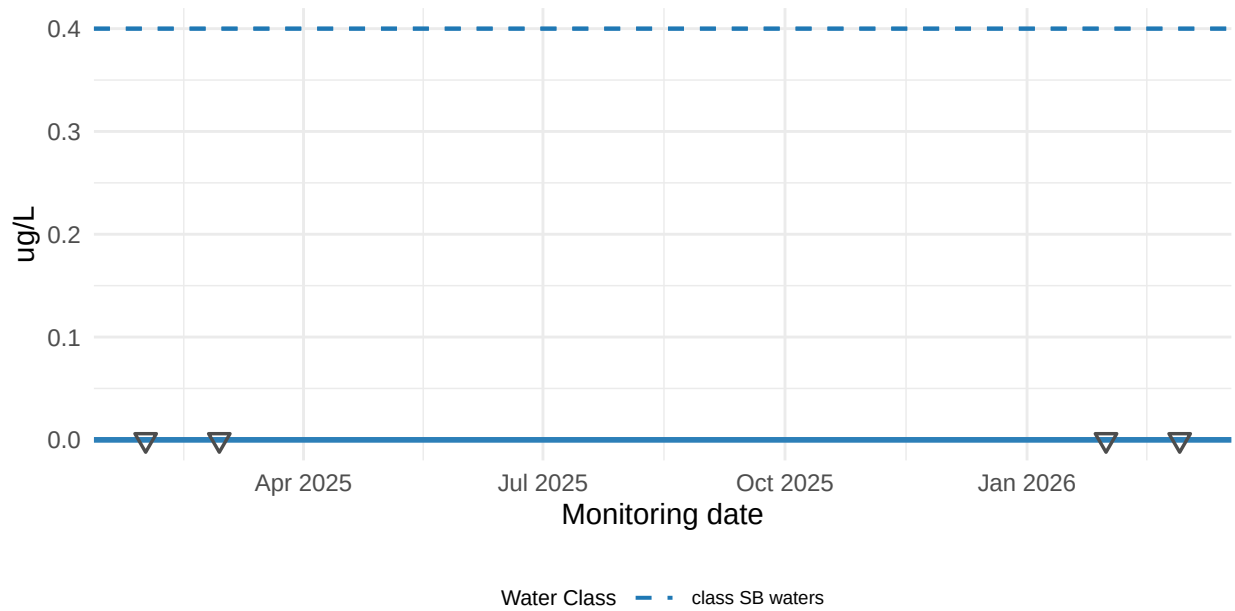
- Samples: 4
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79323	class SB waters	0.4	NO

RWC Calculation

$$\begin{aligned}\sigma_g &= \sqrt{\ln(0.6^2 + 1)} = 0.55 \\ p &= (1 - 0.95)^{1/4} = 0.47 \\ z_p &= -0.068, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - -0.068) \times 0.55) = 2.6 \\ RWC &= \frac{\max(x) \times MF}{DR} = \frac{0 \times 2.6}{1} = 0\end{aligned}$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79323	class SB waters	0.4	0	NO

4.13 Silver

Reason Included: Form(s) 2C selected.

4.13.1 External Outfall 001

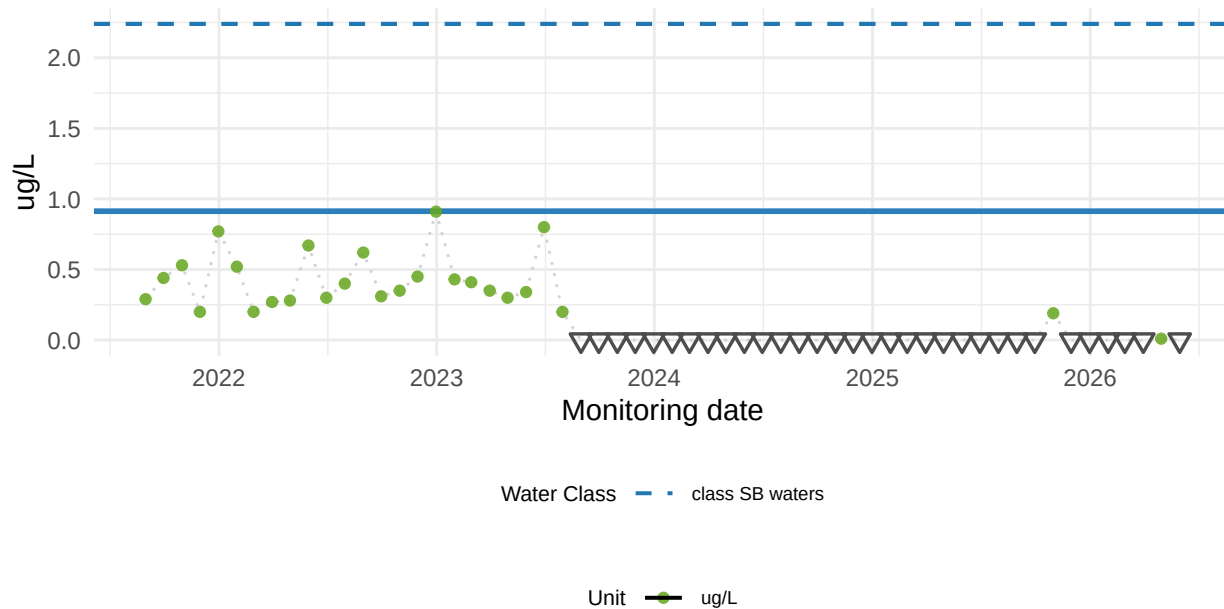
- Samples: 58
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79250	class SB waters	2.24	NO

RWC Calculation

$$\begin{aligned}
 \sigma_g &= \sqrt{\ln(1.4^2 + 1)} = 1 \\
 p &= (1 - 0.95)^{1/58} = 0.95 \\
 z_p &= 1.6, z_{\text{target}} = 1.6 \\
 MF &= \exp((1.6 - 1.6) \times 1) = 1 \\
 \text{RWC} &= \frac{\max(x) \times MF}{DR} = \frac{0.91 \times 1}{1} = 0.91
 \end{aligned}$$



WQS ID	Water Class	WQS Criterion	RWC	RP
79250	class SB waters	2.24	0.913057	NO

4.14 Thallium

Reason Included: Form(s) 2C selected.

4.14.1 External Outfall 001

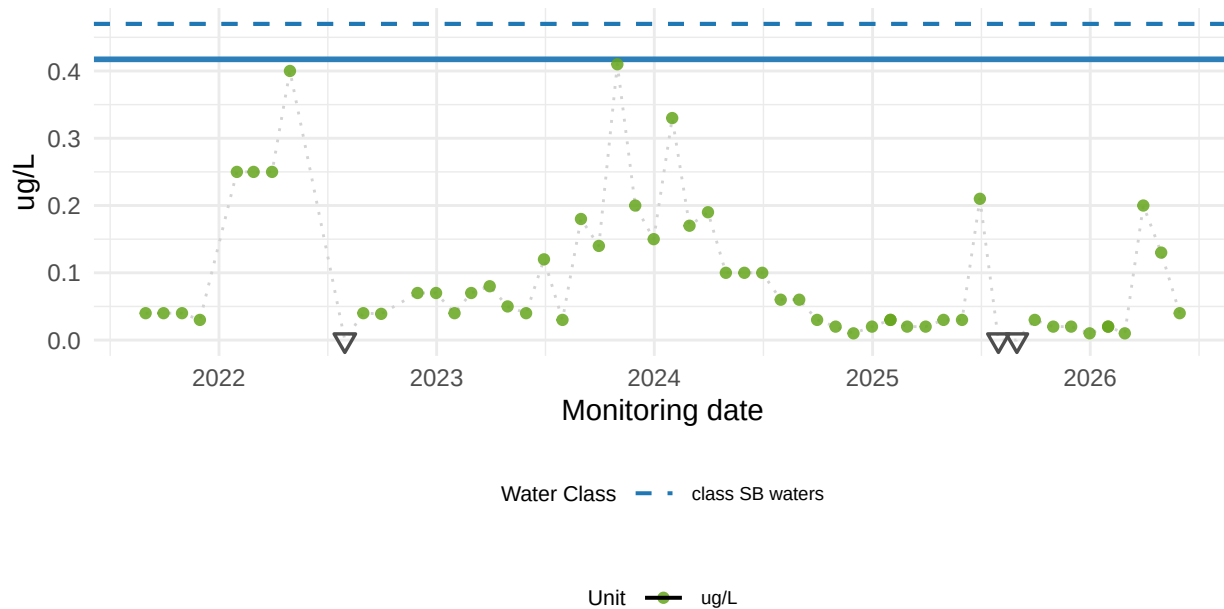
- Samples: 56
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79260	class SB waters	0.47	NO

RWC Calculation

$$\begin{aligned}\sigma_g &= \sqrt{\ln(1.1^2 + 1)} = 0.89 \\ p &= (1 - 0.95)^{1/56} = 0.95 \\ z_p &= 1.6, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - 1.6) \times 0.89) = 1 \\ RWC &= \frac{\max(x) \times MF}{DR} = \frac{0.41 \times 1}{1} = 0.42\end{aligned}$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79260	class SB waters	0.47	0.4173024	NO

4.15 Zinc

Reason Included: Form(s) 2C selected.

4.15.1 External Outfall 01A

- Samples: 30
- Units converted: Yes

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79263	class SB waters	85.62	YES

RWC Calculation

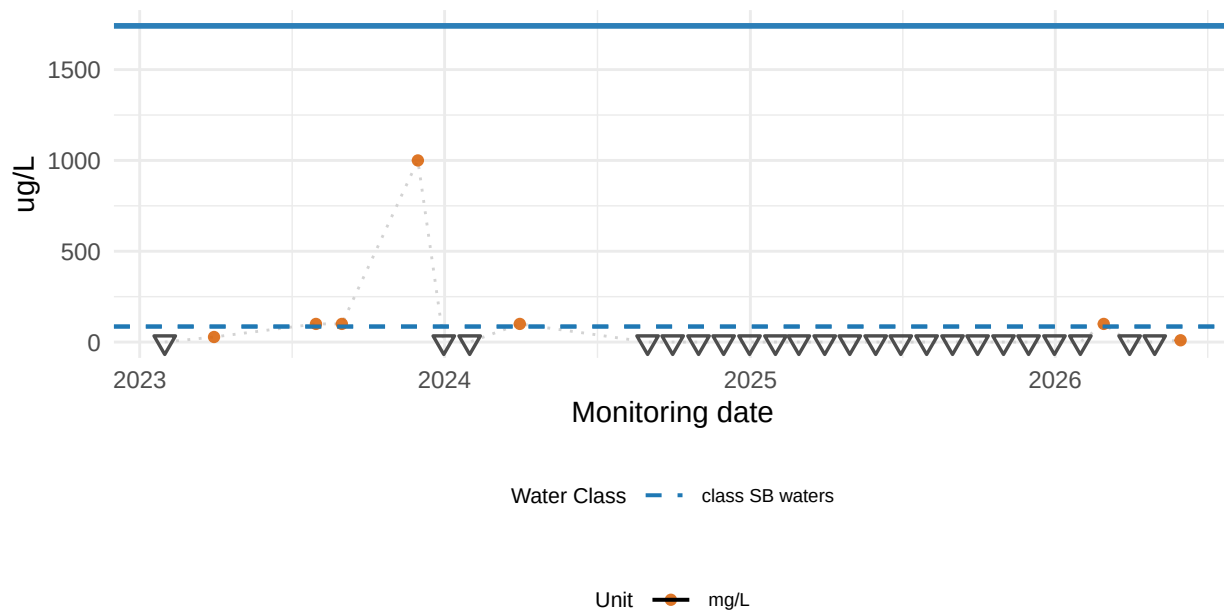
$$\sigma_g = \sqrt{\ln(3.8^2 + 1)} = 1.7$$

$$p = (1 - 0.95)^{1/30} = 0.9$$

$$z_p = 1.3, z_{\text{target}} = 1.6$$

$$MF = \exp((1.6 - 1.3) \times 1.7) = 1.7$$

$$RWC = \frac{\max(x) \times MF}{DR} = \frac{1000 \times 1.7}{1} = 1700$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79263	class SB waters	85.62	1740.732	YES

4.15.2 External Outfall 01C

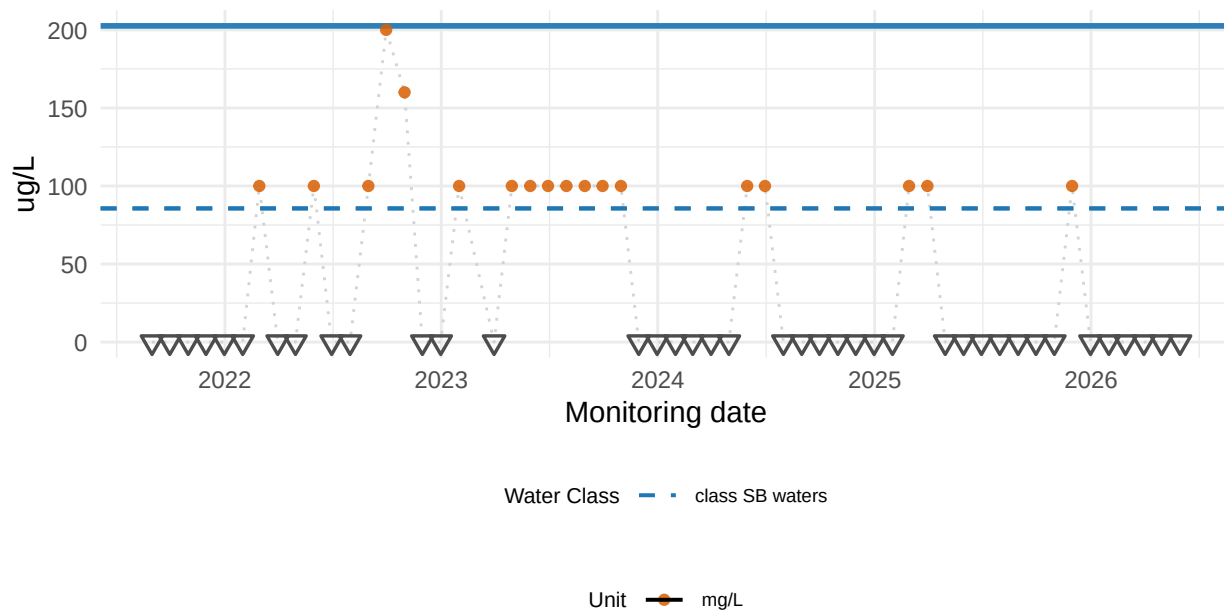
- Samples: 57
- Units converted: Yes

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79263	class SB waters	85.62	YES

RWC Calculation

$$\begin{aligned}\sigma_g &= \sqrt{\ln(1.5^2 + 1)} = 1.1 \\ p &= (1 - 0.95)^{1/57} = 0.95 \\ z_p &= 1.6, z_{\text{target}} = 1.6 \\ \text{MF} &= \exp((1.6 - 1.6) \times 1.1) = 1 \\ \text{RWC} &= \frac{\max(x) \times \text{MF}}{\text{DR}} = \frac{200 \times 1}{1} = 200\end{aligned}$$



Open triangles are non-detects.

WQS ID	Water Class	WQS Criterion	RWC	RP
79263	class SB waters	85.62	202.5619	YES

4.16 Sulfates, total [as SO₄]

Reason Included: Not associated with selected forms — prior DMR data exists.

4.16.1 External Outfall 001

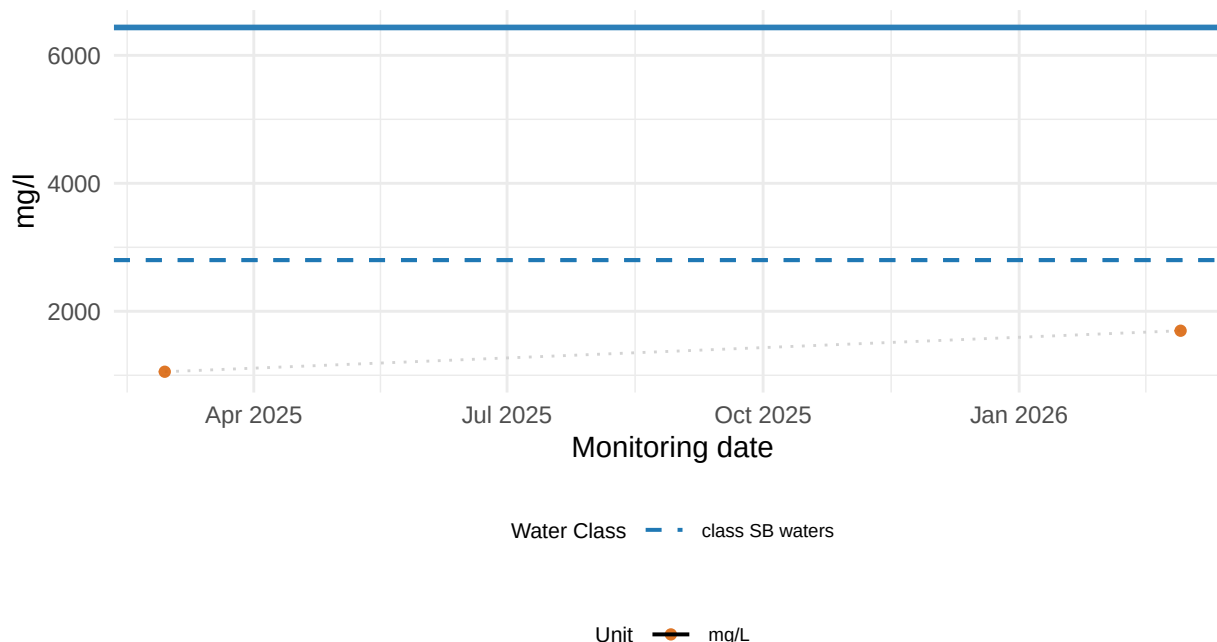
- Samples: 2
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79597-51865	class SB waters	2800	YES

RWC Calculation

$$\begin{aligned}\sigma_g &= \sqrt{\ln(0.6^2 + 1)} = 0.55 \\ p &= (1 - 0.95)^{1/2} = 0.22 \\ z_p &= -0.76, z_{\text{target}} = 1.6 \\ MF &= \exp((1.6 - -0.76) \times 0.55) = 3.8 \\ \text{RWC} &= \frac{\max(x) \times MF}{DR} = \frac{1700 \times 3.8}{1} = 6400\end{aligned}$$



WQS ID	Water Class	WQS Criterion	RWC	RP
79597-51865	class SB waters	2800	6435.524	YES

4.17 Dissolved Oxygen

Note — Dissolved oxygen is evaluated by direct comparison of the observed minimum to the criterion; no RWC is calculated. A defensible DO analysis requires the Streeter-Phelps equation, whose inputs (reaeration and deoxygenation rates, travel time, upstream deficit) are not available in ICIS-NPDES. Where no excursion is observed the determination is DEPENDS, not NO: the permit writer should determine whether additional data must be collected to apply Streeter-Phelps.

4.17.1 External Outfall 001

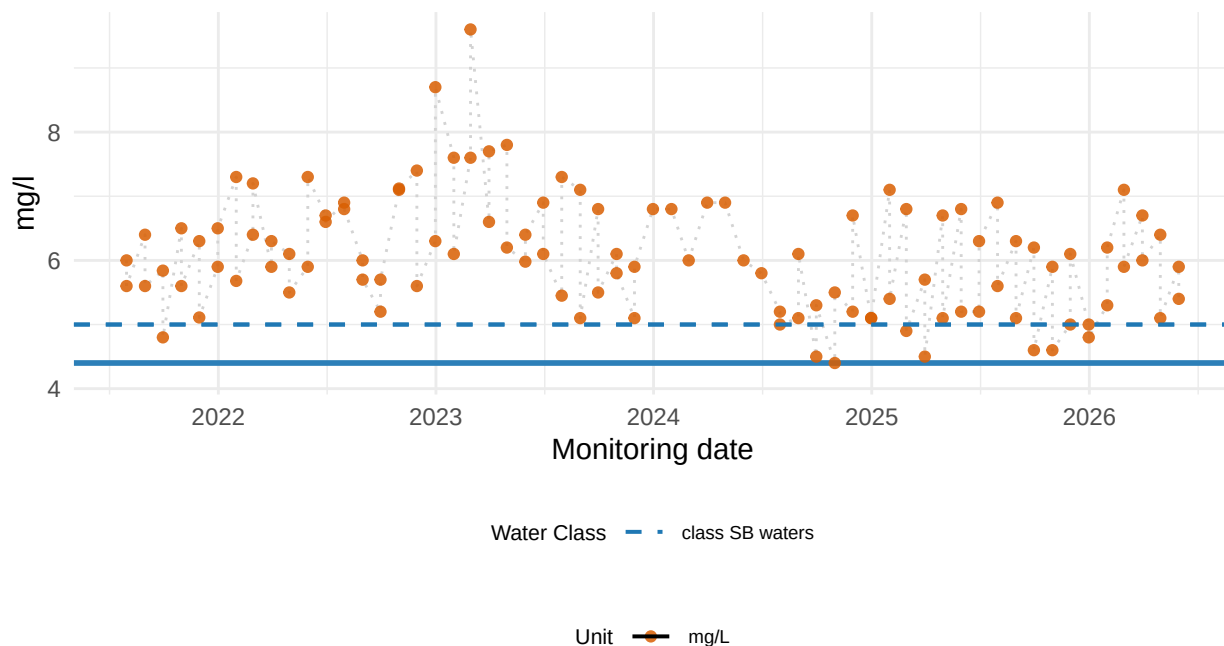
- Samples: 111
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79590	class SB waters	5	YES

Assessed Value

This parameter is assessed by direct comparison. The observed minimum of 4.4 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79590	class SB waters	5	4.4	YES

4.17.2 External Outfall 002

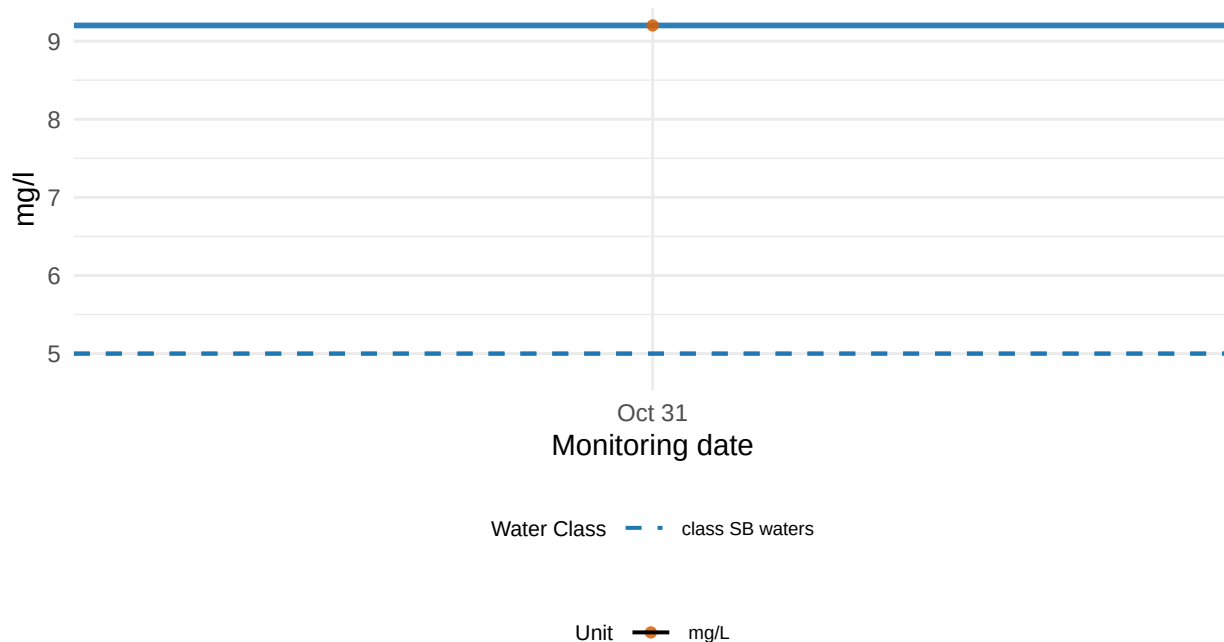
- Samples: 1
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79590	class SB waters	5	DEPENDS

Assessed Value

This parameter is assessed by direct comparison. The observed minimum of 9.2 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79590	class SB waters	5	9.2	DEPENDS

4.18 pH (maximum)

Reason Included: Form(s) 2C & 2F selected.

4.18.1 External Outfall 001

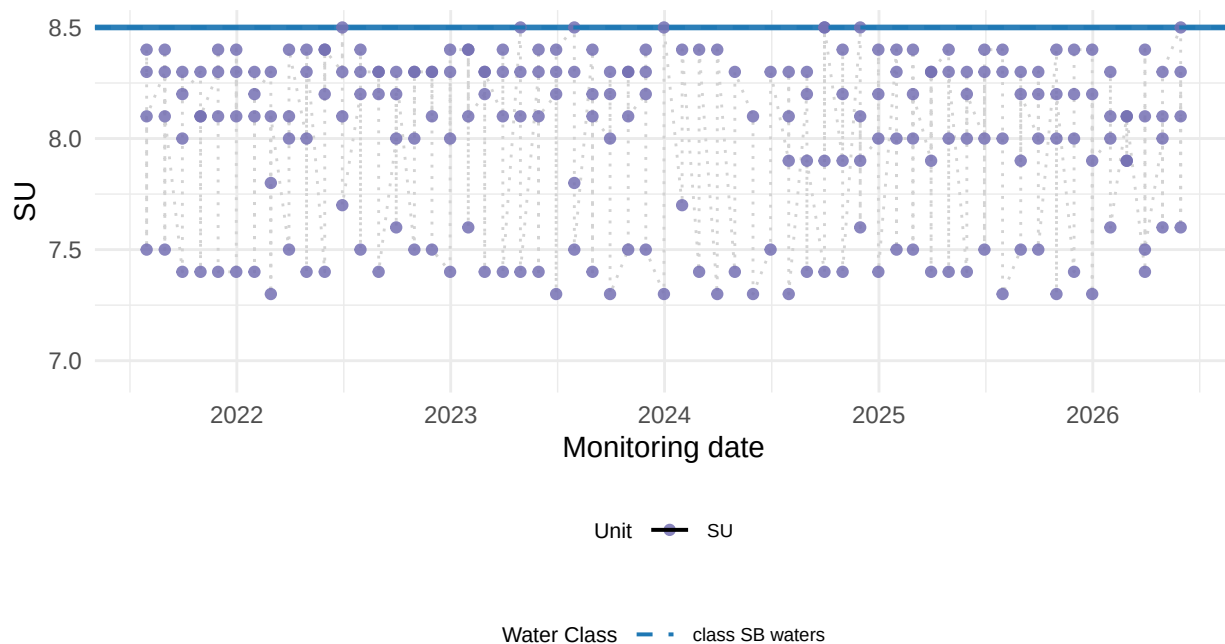
- Samples: 222
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79595	class SB waters	8.5	NO

Assessed Value

This parameter is assessed by direct comparison. The observed maximum of 8.5 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79595	class SB waters	8.5	8.5	NO

4.18.2 External Outfall 002

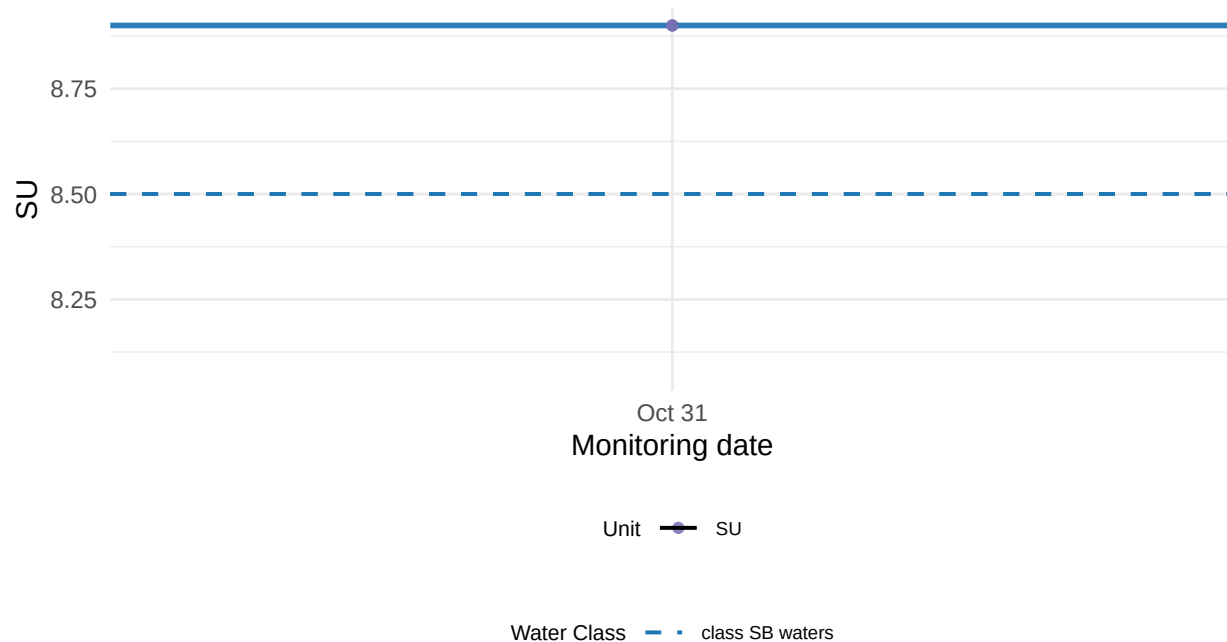
- Samples: 2
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79595	class SB waters	8.5	YES

Assessed Value

This parameter is assessed by direct comparison. The observed maximum of 8.9 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79595	class SB waters	8.5	8.9	YES

4.18.3 External Outfall 01A

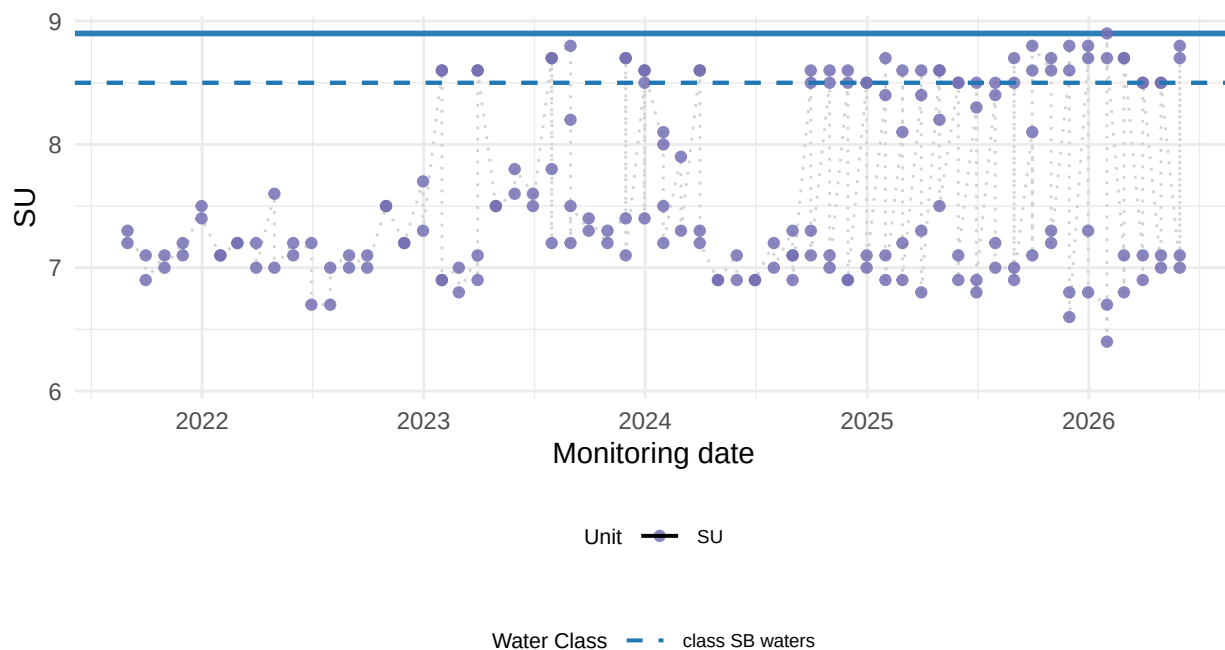
- Samples: 176
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79595	class SB waters	8.5	YES

Assessed Value

This parameter is assessed by direct comparison. The observed maximum of 8.9 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79595	class SB waters	8.5	8.9	YES

4.18.4 External Outfall 01C

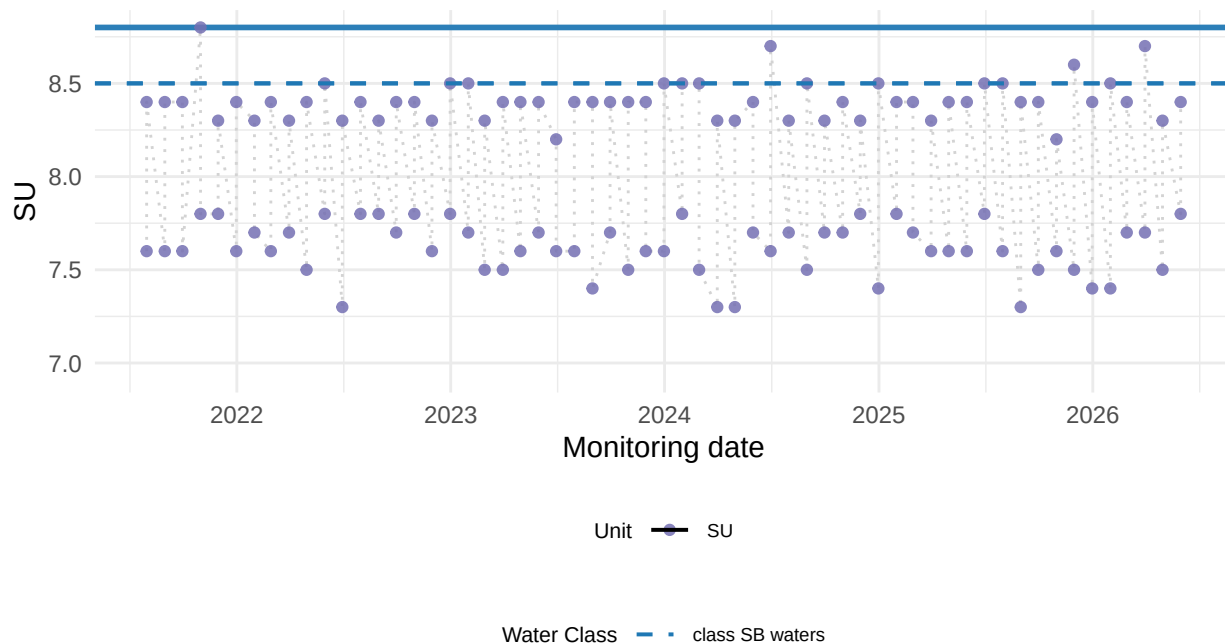
- Samples: 118
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79595	class SB waters	8.5	YES

Assessed Value

This parameter is assessed by direct comparison. The observed maximum of 8.8 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79595	class SB waters	8.5	8.8	YES

4.19 pH (minimum)

Reason Included: Form(s) 2C & 2F selected.

4.19.1 External Outfall 001

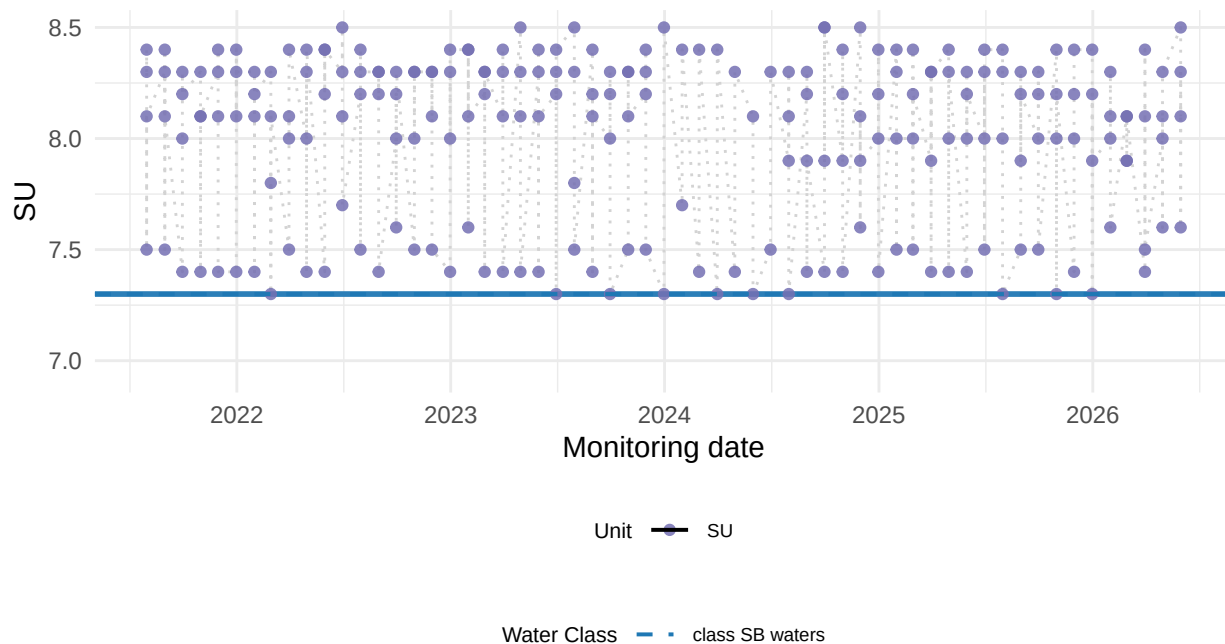
- Samples: 222
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79595	class SB waters	7.3	NO

Assessed Value

This parameter is assessed by direct comparison. The observed minimum of 7.3 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79595	class SB waters	7.3	7.3	NO

4.19.2 External Outfall 002

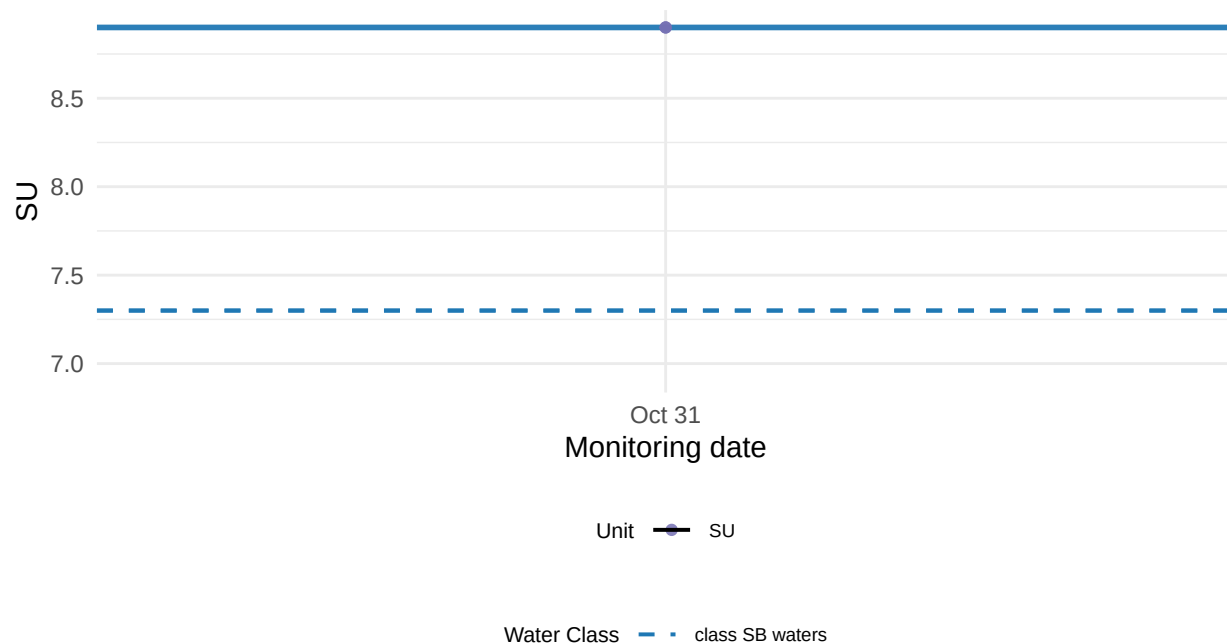
- Samples: 2
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79595	class SB waters	7.3	NO

Assessed Value

This parameter is assessed by direct comparison. The observed minimum of 8.9 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79595	class SB waters	7.3	8.9	NO

4.19.3 External Outfall 01A

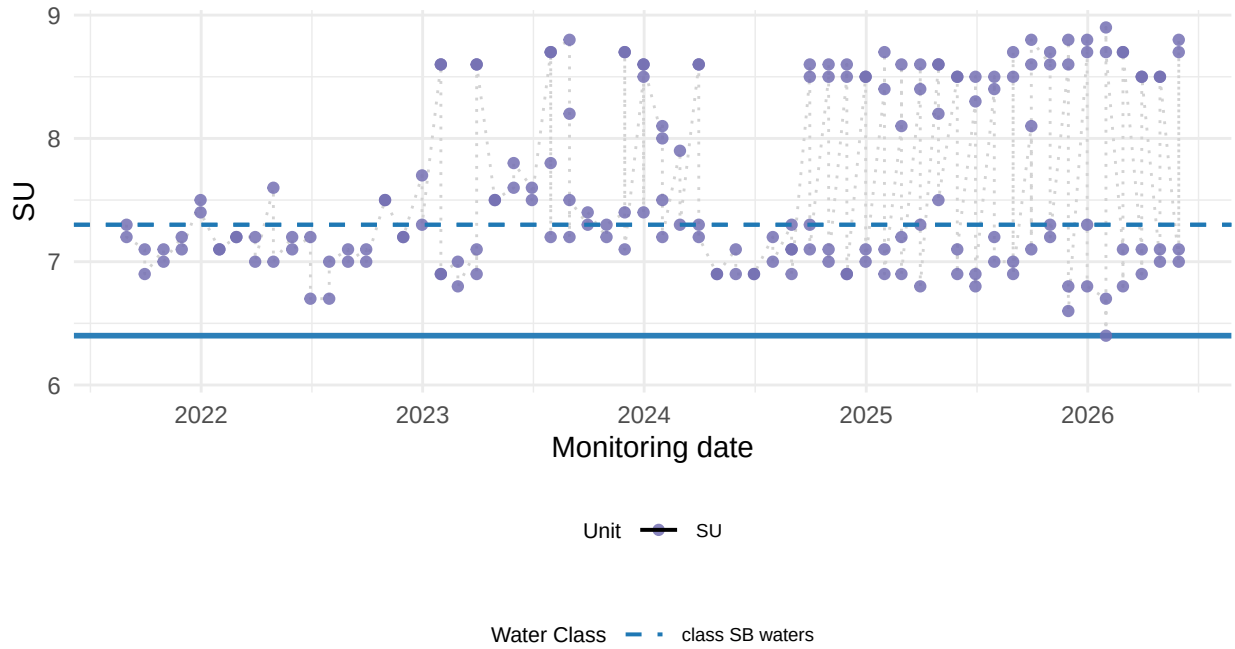
- Samples: 176
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79595	class SB waters	7.3	YES

Assessed Value

This parameter is assessed by direct comparison. The observed minimum of 6.4 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79595	class SB waters	7.3	6.4	YES

4.19.4 External Outfall 01C

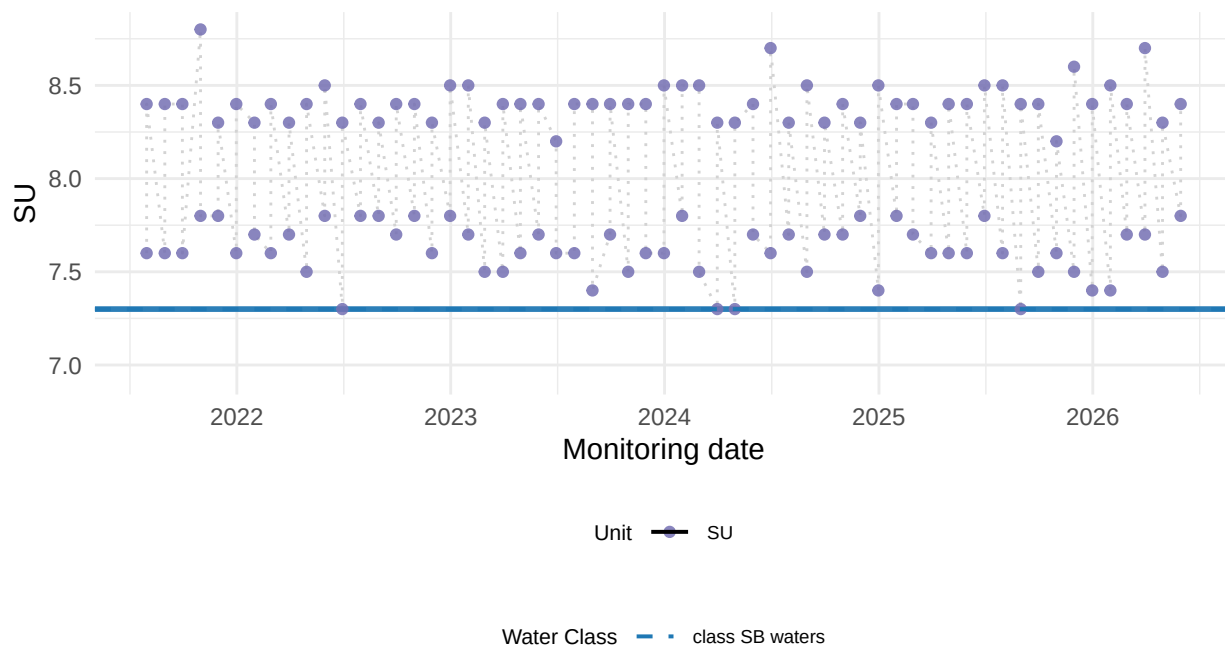
- Samples: 118
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79595	class SB waters	7.3	NO

Assessed Value

This parameter is assessed by direct comparison. The observed minimum of 7.3 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79595	class SB waters	7.3	7.3	NO

4.20 Turbidity

Reason Included: Not associated with selected forms — prior DMR data exists.

4.20.1 External Outfall 001

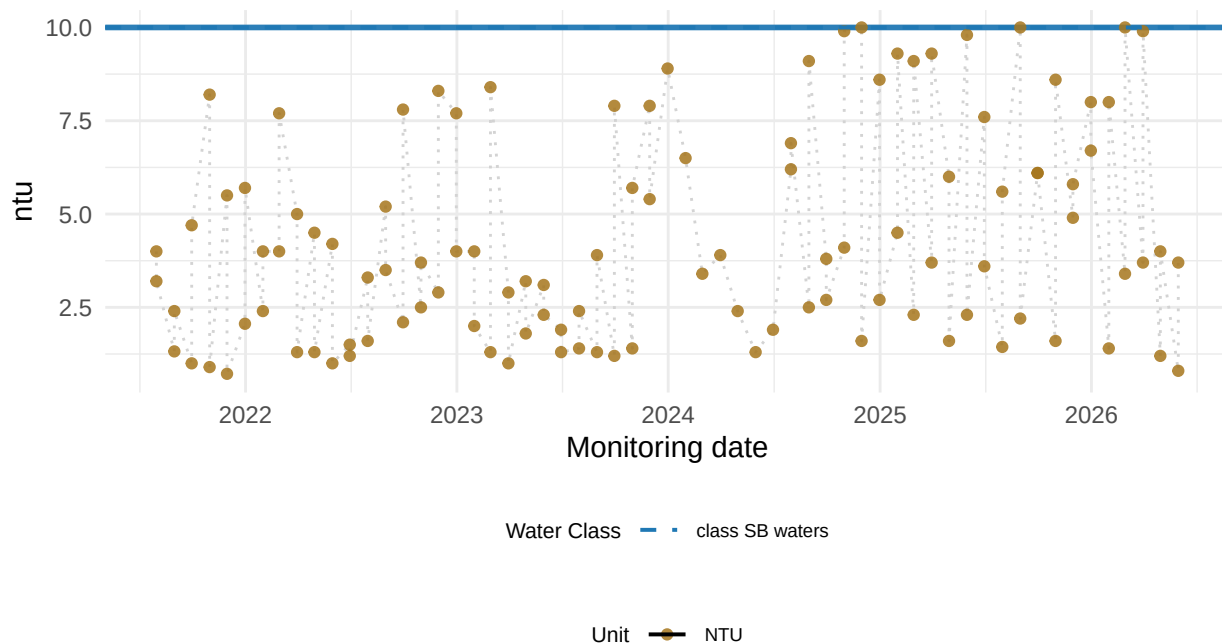
- Samples: 111
- Units converted: No

RP Breakdown by Water Class

WQS ID	Water Class	WQS Criterion	RP
79596	class SB waters	10	NO

Assessed Value

This parameter is assessed by direct comparison. The observed maximum of 10 is compared to the criterion as reported: no multiplying factor is applied and the dilution ratio is not used.



WQS ID	Water Class	WQS Criterion	RWC	RP
79596	class SB waters	10	10	NO

5 Unavailable Pollutants

Pollutant	Ref Param	# Samples
1,1,1-trichloroethane	34506	0
1,1,2,2-tetrachloroethane	34516	0
1,1,2-trichloroethane	34511	0
1,1-dichloroethylene	34501	0
1,2,4-trichlorobenzene	34551	0
1,2-dichlorobenzene	34536	0
1,2-dichloroethane	32103	0
1,2-dichloropropane	34541	0
1,2-diphenylhydrazine	34346	0
1,2-trans-dichloroethylene	34546	0
1,2-trans-dichloroethylene	85810	0
1,3-dichlorobenzene	34566	0
1,3-dichloropropylene	77163	0
1,4-dichlorobenzene	34571	0
2,4-dichlorophenol	34601	0
2,4-dimethylphenol	34606	0
2,4-dinitrophenol	34616	0
2,4-dinitrotoluene	34611	0
2-chloronaphthalene	34581	0
2-chlorophenol	34586	0

(continued)

Pollutant	Ref Param	# Samples
3,3-dichlorobenzidine	34631	0
3,4-benzofluoranthene	34230	0
4,4'-DDD	39310	0
4,4'-DDE	39320	0
4,4'-DDT	39300	0
4,6-dinitro-o-cresol	03615	0
Acenaphthene	34205	0
Acrolein	34210	0
Acrylonitrile	34215	0
Aldrin	39330	0
Anthracene	34220	0
Antimony	01095	0
Antimony	01097	0
Antimony	01268	0
Benzene	34030	0
Benzidine	39120	0
Benzo (a) anthracene	34526	0
Benzo (a) pyrene	34247	0
Benzo (k) fluoranthene	34242	0
Bis (2-chloroethyl) ether	34273	0
Bis (2-ethylhexyl) phthalate	39100	0
Bromoform	32104	0
Butyl benzyl phthalate	34292	0
Carbon tetrachloride	32102	0
Chlordane	51032	0
Chlorobenzene	34301	0
Chlorodibromomethane	34306	0
Chloroform	32106	0
Chrysene	34320	0
Di-n-butyl phthalate	51011	0
Dibenzo (a,h) anthracene	34556	0
Dichlorobromomethane	32101	0
Dieldrin	39380	0
Diethyl phthalate	34336	0
Dimethyl phthalate	34341	0
Dimethyl phthalate	34342	0
Endosulfan sulfate	34351	0
Endrin	39390	0
Endrin aldehyde	34366	0
Ethylbenzene	34371	0
Ethylbenzene	37371	0
Ethylbenzene	78113	0
Fluoranthene	34376	0
Fluorene	34381	0
Heptachlor	39410	0
Heptachlor epoxide	39420	0
Hexachlorobenzene	39700	0

(continued)

Pollutant	Ref Param	# Samples
Hexachlorobutadiene	34391	0
Hexachlorobutadiene	39702	0
Hexachlorocyclopentadiene	34386	0
Hexachloroethane	34396	0
Indeno (1,2,3-cd) pyrene	34403	0
Isophorone	34408	0
Lead	01051	0
Methyl bromide	34413	0
Methylene chloride	34423	0
N-nitrosodi-n-propylamine	34428	0
N-nitrosodimethylamine	34438	0
N-nitrosodiphenylamine	34433	0
Nitrobenzene	34447	0
Phenol	34694	0
Pyrene	34469	0
Selenium	01147	0
Surfactants	47021	0
Surfactants	38260	0
Tetrachloroethylene	34475	0
Tetrachloroethylene	85814	0
Toluene	34010	0
Toluene	34481	0
Total nitrogen (as N)	00600	0
Total phosphorus	00665	0
Toxaphene	39400	0
Trichloroethylene	34485	0
Trichloroethylene	39180	0
Vinyl chloride	39175	0
alpha-BHC	39336	0
alpha-BHC	39337	0
beta-BHC	34255	0
beta-BHC	39338	0
beta-BHC	51007	0
gamma-BHC	39782	0
p-chloro-m-cresol	70012	0

6 Excluded Parameters

The following parameters had DMR data available but were excluded from the RP analysis at the direction of the user.

Parameter	Code	Flag Reason	Reason for Exclusion
Temperature, water deg. centigrade	00010	Wrong water class	Auto-excluded: parameter not reviewed prior to Quick Run
Solids, settleable	00545	No crosswalk entry	Auto-excluded: parameter not reviewed prior to Quick Run
Flow, in conduit or thru treatment plant	50050	No crosswalk entry	Auto-excluded: parameter not reviewed prior to Quick Run

(continued)

Parameter	Code	Flag Reason	Reason for Exclusion
BOD, 5-day, 20 deg. C	00310	No crosswalk entry	Auto-excluded: parameter not reviewed prior to Quick Run
Solids, total suspended	00530	No crosswalk entry	Auto-excluded: parameter not reviewed prior to Quick Run
Oil & Grease	00556	Wrong water class	Auto-excluded: parameter not reviewed prior to Quick Run
Chromium, total [as Cr]	01034	Wrong water class	Auto-excluded: parameter not reviewed prior to Quick Run
Iron, total [as Fe]	01045	No crosswalk entry	Auto-excluded: parameter not reviewed prior to Quick Run
Oxygen demand, chem. [high level] [COD]	00340	No crosswalk entry	Auto-excluded: parameter not reviewed prior to Quick Run
Chlorine, free available	50064	No crosswalk entry	Auto-excluded: parameter not reviewed prior to Quick Run
Nitrogen, Total	51445	No crosswalk entry	Auto-excluded: parameter not reviewed prior to Quick Run

7 Manual WQS Overrides

No manual WQS criteria were entered for this analysis.